DATA & BI ANALYTICS

Career Roadmap

Basic → Intermediate

The definitive 2026 guide for Junior & Mid-Level Analysts

Ideal for: Freshers | Career Switchers | 0–4 Years Experience

7 Core Domains	25+ Tools Covered	4-6 Month Path	Industry-Verified

How to Use This Roadmap

1. Start with Phase 0 (Mindset & Foundations) before touching any tool.
2. Follow phases sequentially — each phase builds on the previous one.
3. Use the Priority column: Must-Have > High > Good-to-Have.
4. Build at least one real project per phase and push it to GitHub / portfolio.
5. Revisit and deepen skills while advancing — analytics is iterative by nature.

01. What Is Data Analytics Today?

Data Analytics in 2026 is far beyond Excel and bar charts. It sits at the intersection of business strategy, engineering, and statistical science. Companies now expect analysts to own the full data lifecycle — from querying raw data to delivering actionable insights through dashboards, predictive models, and storytelling.

What Analysts Do

- Define business questions and KPIs
- Clean, transform, and validate data
- Explore data through EDA
- Build dashboards and automated reports
- Apply statistical models and A/B tests
- Communicate findings to stakeholders
- Collaborate with Data Engineers & Scientists

Where Analysts Work

- E-Commerce & Retail (Amazon, Flipkart)
- FinTech & Banking (Paytm, HDFC, Zerodha)
- Healthcare & Pharma (Apollo, Practo)
- EdTech (Byju's, Unacademy, Coursera)
- Consulting (Deloitte, McKinsey, PwC)
- Startups & SaaS companies
- Government & Public Sector

Analyst Career Ladder

Role	Experience	Key Responsibility	India CTC Range
Data / Business Analyst Intern	0–6 months	Reporting, dashboards, ad-hoc queries	3–6 LPA (stipend)
Junior Data Analyst	**0–2 years**	**SQL, Excel, basic dashboards, EDA**	**4–8 LPA**
Data Analyst	2–4 years (THIS ROADMAP)	End-to-end analysis, Python, BI tools	8–15 LPA
Senior Data Analyst	4–7 years	Strategy, mentoring, advanced models	15–25 LPA
Lead / Principal Analyst	7+ years	Cross-functional ownership, AI/ML integration	25–40 LPA

★ Highlighted row = target level of this roadmap

PHASE 0 | FOUNDATION

Mindset & Setup

Before writing a single query — build the right mental model

Core Principles Every Analyst Must Internalize

🧠 The Analyst Mindset

✓ Data is a means, not the end — the end is the decision it enables.

✓ Question the question — always clarify what business problem you are solving before analyzing.

✓ Correlation ≠ Causation — never draw causal conclusions without rigorous methodology.

✓ Reproducibility matters — every analysis must be repeatable by someone else.

✓ Communication > Complexity — a simple insight delivered clearly beats a complex model explained poorly.

✓ Version control everything — use Git from day one, no exceptions.

✓ Documentation is part of the job — undocumented analysis has no business value.

Environment Setup Checklist

Skill / Topic	Level	Tools / Technologies	Priority
Git & GitHub	**Basic**	Git CLI, GitHub / GitLab	**Must-Have**
VS Code / IDE	**Basic**	VS Code, Jupyter Notebook, Google Colab	**Must-Have**
Python Setup	**Basic**	Anaconda, pip, virtual environments	**Must-Have**
Cloud Account	**Basic**	Google Cloud (free tier), AWS Free Tier	**High**
Linux Basics	**Basic**	WSL2 (Windows), Terminal (Mac/Linux)	**High**
Markdown	**Basic**	VS Code, Obsidian, Notion	**High**

PHASE 1 | CORE FOUNDATIONS
Mathematics, Statistics & Excel
The non-negotiable bedrock of every analyst's career

1.1 Mathematics & Statistics

Statistics is the language of data analytics. Without a solid statistical foundation, you cannot trust your analyses, identify real patterns, or make defensible recommendations.

Skill / Topic	Level	Tools / Technologies	Priority
Descriptive Statistics	Basic	Excel, Python (pandas, numpy)	Must-Have
Probability Foundations	Basic	Python, Khan Academy, StatQuest	Must-Have
Normal Distribution	Basic	scipy.stats, seaborn	Must-Have
Hypothesis Testing	Intermediate	scipy, statsmodels, Python	Must-Have
Confidence Intervals	Intermediate	scipy, statsmodels	Must-Have
A/B Testing	Intermediate	Python, Optimizely concepts	Must-Have
Correlation & Regression	Intermediate	sklearn, statsmodels, Excel	High
Central Limit Theorem	Basic	Python simulations	High
Bayes Theorem Basics	Intermediate	Python, textbooks	Good-to-Have

1.2 Excel / Google Sheets — Still Essential

Despite the rise of Python and BI tools, Excel remains the universal language of business. You WILL need it for quick analysis, client deliverables, and ad-hoc requests.

Skill / Topic	Level	Tools / Technologies	Priority
VLOOKUP / XLOOKUP / INDEX-MATCH	Basic	Excel, Google Sheets	Must-Have
Pivot Tables & Charts	Basic	Excel, Google Sheets	Must-Have

Skill / Topic	Level	Tools / Technologies	Priority
Conditional Formatting	Basic	Excel	Must-Have
Power Query (ETL in Excel)	Intermediate	Excel Power Query	High
Array Formulas / SUMPRODUCT	Intermediate	Excel	High
Excel Macros (VBA basics)	Intermediate	Excel VBA	Good-to-Have
Google Sheets + Apps Script	Intermediate	Google Workspace	Good-to-Have

PHASE 2 | THE ANALYST'S PRIMARY LANGUAGE

SQL & Databases

SQL is the single most-tested skill in every data analytics interview

2.1 SQL — From Basic to Intermediate

SQL is non-negotiable. Every data analyst role on the planet requires SQL. Master it before everything else. Most companies use PostgreSQL, BigQuery, Snowflake, or Redshift.

Skill / Topic	Level	Tools / Technologies	Priority
SELECT, WHERE, ORDER BY	Basic	PostgreSQL, MySQL, SQLite	Must-Have
GROUP BY & Aggregations	Basic	Any SQL engine	Must-Have
JOINs (INNER, LEFT, RIGHT, FULL)	Basic	PostgreSQL, BigQuery	Must-Have
Subqueries & CTEs (WITH clause)	Intermediate	PostgreSQL, Snowflake, BigQuery	Must-Have
Window Functions (RANK, LAG, LEAD, ROW_NUMBER)	Intermediate	BigQuery, Snowflake, PostgreSQL	Must-Have
CASE WHEN & Conditional Logic	Basic	Any SQL engine	Must-Have
Date/Time Functions	Basic	PostgreSQL, BigQuery, MySQL	Must-Have
String Functions (CONCAT, LIKE, REGEX)	Intermediate	PostgreSQL, BigQuery	High
Indexes & Query Optimization	Intermediate	PostgreSQL, EXPLAIN ANALYZE	High
Cloud SQL (BigQuery / Redshift)	Intermediate	Google BigQuery, AWS Redshift	High

Skill / Topic	Level	Tools / Technologies	Priority
Stored Procedures & Views	Intermediate	PostgreSQL, SQL Server	Good-to-Have
Database Design & Normalization	Intermediate	dbdiagram.io, ERD tools	Good-to-Have

📋 SQL Practice Platforms

→ LeetCode (SQL section) — Interview-style problems, free tier available
→ Mode Analytics SQL Tutorial — Real business datasets, excellent for beginners
→ StrataScratch — Company-tagged interview questions (Google, Amazon, Facebook)
→ SQLZoo — Interactive exercises with immediate feedback
→ HackerRank SQL — Competitive, good for building speed
→ BigQuery Sandbox (free) — Practice on real cloud infrastructure at zero cost

PHASE 3 | AUTOMATION & ANALYSIS AT SCALE

Python for Analytics

From data wrangling to predictive modelling — Python is your analyst superpower

3.1 Python Fundamentals

You do not need to be a software engineer. Focus on using Python as an analytical tool — not building apps. The key libraries are pandas, numpy, matplotlib, seaborn, and scikit-learn.

Skill / Topic	Level	Tools / Technologies	Priority
Python Syntax & Data Types	Basic	Python 3.x, Jupyter Notebook	Must-Have
Lists, Dicts, Loops, Functions	Basic	Python, Colab	Must-Have
File I/O (CSV, JSON, Excel)	Basic	Python, pandas	Must-Have
pandas — DataFrame Operations	Basic	pandas	Must-Have
pandas — merge, groupby, pivot_table	Intermediate	pandas	Must-Have
numpy — Arrays & Vectorization	Basic	numpy	Must-Have
Data Cleaning (nulls, types, duplicates)	Intermediate	pandas, pyjanitor	Must-Have
EDA — Exploratory Data Analysis	Intermediate	pandas, ydata-profiling	Must-Have
API Data Fetching	Intermediate	requests, json	High
Web Scraping Basics	Intermediate	BeautifulSoup, Selenium	Good-to-Have

3.2 Data Visualization with Python

Skill / Topic	Level	Tools / Technologies	Priority
Matplotlib — Base Plotting	Basic	matplotlib	Must-Have
Seaborn — Statistical Plots	Basic	seaborn	Must-Have
Plotly — Interactive Charts	Intermediate	plotly, plotly.express	High
Dashboard in Python (Streamlit)	Intermediate	Streamlit, Dash	High
Visualization Best Practices	Intermediate	Storytelling with Data (book)	Must-Have

3.3 Statistics with Python

Skill / Topic	Level	Tools / Technologies	Priority
Descriptive Stats with pandas/numpy	Basic	pandas, numpy, scipy	Must-Have
Hypothesis Testing in Python	Intermediate	scipy.stats, statsmodels	Must-Have
Linear Regression (OLS)	Intermediate	statsmodels, sklearn	Must-Have
Logistic Regression	Intermediate	sklearn, statsmodels	High
Time Series Analysis Basics	Intermediate	pandas, statsmodels (ARIMA)	High
Clustering Basics (K-Means)	Intermediate	sklearn	Good-to-Have

PHASE 4 | INSIGHTS AT THE SPEED OF BUSINESS

Business Intelligence & Dashboards

Translate complex data into decisions stakeholders can act on immediately

4.1 Business Intelligence Tools

BI tools are how most organizations consume data insights. You need to master at least ONE tool deeply. Power BI dominates enterprise India; Tableau leads globally; Looker is the choice of data-mature tech companies.

Skill / Topic	Level	Tools / Technologies	Priority
Power BI Desktop	Basic	Power BI Desktop (free)	Must-Have
DAX — Calculated Columns & Measures	Intermediate	Power BI DAX	Must-Have
Power BI Service (Publish & Share)	Intermediate	Power BI Pro / Premium	High
Tableau Public / Desktop	Basic	Tableau Public (free)	Must-Have
Tableau Calculated Fields & LOD	Intermediate	Tableau Desktop	High
Looker Studio (ex-Data Studio)	Basic	Google Looker Studio (free)	High
Looker (LookML)	Intermediate	Looker / Google Cloud	Good-to-Have
Dashboard Design Principles	Intermediate	Any BI tool + design thinking	Must-Have
KPI Tracking & Metrics Layer	Intermediate	Power BI, dbt Metrics, Looker	High

🎨 Dashboard Design Principles (Must-Know)

→ Single source of truth: every metric must trace back to one definition.
→ 5-second rule: a stakeholder should grasp the main insight in 5 seconds.

→ Hierarchy of attention: most important KPI top-left; supporting charts below.
→ Use color intentionally: red = bad, green = good, blue = neutral — be consistent.
→ Avoid 3D charts, pie charts with >5 slices, and dual-axis charts without labels.
→ Always include context: YoY, MoM comparison, benchmarks, targets.
→ Mobile-first is now standard: ensure dashboards work on tablet/phone screens.

PHASE 5 | KNOW WHERE DATA COMES FROM

Data Engineering Fundamentals

Analysts who understand pipelines are 2x more effective than those who don't

5.1 Data Engineering for Analysts

You are not expected to build production pipelines — that is the Data Engineer's job. But you must understand how data flows, where it lives, and why it might be wrong.

Skill / Topic	Level	Tools / Technologies	Priority
ETL vs ELT Concepts	Basic	Theory + dbt, Fivetran	Must-Have
Data Warehouse Concepts	Basic	BigQuery, Snowflake, Redshift	Must-Have
Data Lake vs Data Warehouse	Basic	Conceptual + AWS S3, GCS	Must-Have
dbt (Data Build Tool)	Intermediate	dbt Core / Cloud (free)	High
Airflow Basics (DAGs)	Intermediate	Apache Airflow, Astronomer	Good-to-Have
Fivetran / Airbyte (Connectors)	Basic	Fivetran, Airbyte (open-source)	Good-to-Have
Data Quality & Validation	Intermediate	Great Expectations, dbt tests	High
Schema Design (Star vs Snowflake)	Intermediate	BigQuery, Snowflake, LucidChart	High
Streaming Data Basics	Basic	Kafka concepts, Pub/Sub	Good-to-Have

5.2 Cloud Platforms for Analysts

Skill / Topic	Level	Tools / Technologies	Priority
Google BigQuery (SQL on Cloud)	Intermediate	BigQuery Sandbox (free 10GB/month)	Must-Have
Google Cloud Storage	Basic	GCS Free Tier	High
AWS S3 + Athena	Basic	AWS Free Tier	High
Snowflake (Cloud DW)	Intermediate	Snowflake Free Trial	High
Azure Synapse Basics	Basic	Azure Free Account	Good-to-Have

PHASE 6 | THE MULTIPLIER SKILLS

Storytelling, Communication & Domain Knowledge

Technical skills get you the interview — these skills get you promoted

6.1 Data Storytelling & Communication

Skill / Topic	Level	Tools / Technologies	Priority
Structuring an Analysis Narrative	Intermediate	SCQA Framework, Pyramid Principle	Must-Have
Executive Slide Decks	Intermediate	PowerPoint, Google Slides, Canva	Must-Have
Annotation & Insight Callouts	Intermediate	Tableau, Power BI, Figma	High
Writing Analysis Reports	Intermediate	Notion, Confluence, Google Docs	High
Presenting to Non-Technical Stakeholders	Intermediate	Practice, Toastmasters	Must-Have
Asking the Right Business Questions	Intermediate	5-Whys, Jobs-to-be-Done framework	Must-Have

6.2 Domain Knowledge (Choose at Least One)

Deep domain knowledge separates average analysts from excellent ones. Choose a domain aligned with your target industry and master its metrics, terminology, and decision cycles.

Domain	Key Metrics to Know	Tools / Sources
E-Commerce	CAC, LTV, Conversion Rate, Churn, GMV, AOV	GA4, Mixpanel, Shopify Analytics
FinTech / BFSI	NPA, AUM, Default Rate, ARPU, Credit Score Dist.	RBI reports, Bloomberg, company filings
Marketing Analytics	ROAS, CPL, CTR, Funnel Conversion, Attribution	Google Ads, Meta Ads, GA4, Clevertap

Domain	Key Metrics to Know	Tools / Sources
Product Analytics	DAU/MAU, Retention, NPS, Feature Adoption	Mixpanel, Amplitude, FullStory
Healthcare	Readmission Rate, Length of Stay, Cost per Claim	HMIS, HL7, government health data
SaaS / Subscription	MRR, ARR, Churn, NRR, CAC Payback Period	Stripe, ChartMogul, ProfitWell

02. Recommended 3-6 Month Learning Timeline

This is a suggested pace for someone studying 2–3 hours daily. Adjust based on your prior experience. Career switchers with programming background can compress; absolute beginners should expand.

Projects to Build Per Phase

Phase 1–2 Project: Sales Dashboard — Excel + SQL (retail or e-commerce dataset)
Phase 3 Project: Customer Churn EDA — Python (pandas, seaborn, hypothesis testing)
Phase 4 Project: Marketing Analytics Dashboard — Power BI or Tableau (GA4 or social data)
Phase 5 Project: End-to-End Pipeline — BigQuery + dbt + Looker Studio
Final Portfolio: 3+ projects on GitHub, 1 Kaggle notebook, 1 blog post on Medium / Substack

EMERGING SKILLS | 2026 & BEYOND

AI-Era Analytics

The analyst who ignores AI will be replaced by the analyst who uses AI

AI & Automation Skills for Modern Analysts

In 2026, AI tools are transforming analytics workflows. You are not expected to build LLMs — but you must know how to use them to multiply your productivity.

Skill / Topic	Level	Tools / Technologies	Priority
Prompt Engineering for Analytics	Basic	ChatGPT, Claude, Gemini	Must-Have
AI-Assisted SQL Writing	Basic	GitHub Copilot, ChatGPT, SQLChat	Must-Have
Python Copilot (Code Assist)	Basic	GitHub Copilot, Cursor IDE	High
AI-Powered EDA Tools	Intermediate	Julius AI, Pandas AI, Marimo	High
Natural Language to SQL (NL2SQL)	Intermediate	MindsDB, Defog, Vanna.ai	High
LLM for Data Summarization	Intermediate	LangChain, OpenAI API	Good-to-Have
AutoML for Analyst Predictions	Intermediate	BigQuery AutoML, H2O.ai, DataRobot	Good-to-Have
Semantic Layer & Metrics	Intermediate	dbt Semantic Layer, Cube.dev	High

Upcoming Tools to Watch

🔧 Tools Gaining Adoption (2026)

- Motherduck — DuckDB in the cloud
- MotherDuck + dbt — lightweight modern stack
- Evidence.dev — Code-first BI
- Rill Data — Operational dashboards

📈 Skills with Highest Demand Growth

- dbt + Snowflake/BigQuery stack
- Python + Streamlit for internal tools
- A/B Testing & Causal Inference
- NLP for customer feedback analysis
- Real-time analytics (Kafka + Flink)

- Hex — Collaborative notebooks
- Observable — JavaScript data notebooks
- dbt Core 1.8+ — Semantic models

- LLM integration in analytics workflows
- Data governance & lineage tracking

05. Job-Ready Checklist

Before applying for Data Analyst roles, verify each item below. This is what hiring managers at top companies look for:

	Milestone	Evidence / Artifact
☐	Write 15+ intermediate SQL queries without reference	StrataScratch / LeetCode profile
☐	Complete end-to-end Python EDA on a real dataset	Kaggle notebook / GitHub repo
☐	Build and publish a dashboard (Power BI or Tableau)	Tableau Public / Power BI shared link
☐	Conduct an A/B test with statistical significance check	Jupyter notebook on GitHub
☐	Understand 5+ statistical concepts well enough to explain in 1 min	Practice with a friend
☐	Have 3 portfolio projects on GitHub with README files	github.com/yourusername
☐	Write a data analysis blog post (Medium / Substack / LinkedIn)	Article URL
☐	Know at least one cloud platform (BigQuery or Redshift)	Free sandbox projects
☐	Complete mock interview (SQL + case study + behavioral)	Pramp, Exponent, or a peer
☐	Optimize LinkedIn: headline, skills, featured projects	linkedin.com/in/yourprofile
☐	Tailor resume to specific JDs with metrics-driven bullets	PDF resume (1 page)
☐	Apply to 50+ roles and track in a spreadsheet	Job tracker sheet

"Data will talk to you if you're willing to listen."

— Jim Bergeson

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